

Multi-Class Classification of Medicinal Plant Leaf using Machine Learning Approach

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Abstract: The traditional use of medicinal herbs in dermatology offers effective natural remedies for various skin conditions. This work applies machine learning algorithms to classify four medicinal plants like Neem (*Azadirachta indica*), Aloe Vera (*Aloe barbadensis*), Papaya (*Carica papaya*), and Pumpkin (*Cucurbita*) from leaf images and predict their dermatological applications. A dataset comprising 2,000 healthy leaf images (500 per plant) was used to train and evaluate Support Vector Machine (SVM), Random Forest (RF), Decision Tree (DT), Logistic Regression (LR), and K-Nearest Neighbor (KNN) models. Among which Random Forest achieved the highest accuracy of 92.75%, outperforming the other algorithms. The proposed botanical based dermatological decision support system accurately identifies plant species and highlights the skin care benefits, including acne treatment, burn relief and skin radiance enhancement.

Keywords: Medicinal Plant, Skin Care Treatment, Support Vector Machine, Random forest, Logistic regression, Decision tree, K-Nearest Neighbor

I. INTRODUCTION

Medicinal herbs used for beneficial properties and natural origins, which make them play a vital role in both ancient and modern medical systems. Diseases that affect their leaves, however, can have a major effect on these plants' quality and productivity. To tackle this, scientists have shown how well image processing methods can detect and categorize leaf diseases[1]. A hybrid approach combining Convolutional Neural Networks, and an enhanced SVM was developed to enhance the classification of medicinal plants using leaf image data. This method demonstrates how feature learning and traditional algorithms can be combined for increased accuracy[2]. The relevance of computer vision in safeguarding indigenous medicinal resources was highlighted and it focused on Ayurvedic plant preservation and created a machine learning pipeline to identify illnesses in leaves. Their research emphasizes the value of digital tools in preserving the health of plants that hold cultural significance[3]. The cultivation of medicinal plants is influenced by environmental conditions. The use of ML to map soil suitability and assist farmers to identify the best

places to cultivate the particular medicinal species. This regional analysis guarantees increased yield and resilience to stressors associated with climate change[4]. Furthermore, recent studies have reviewed the ethnobotanical significance of several species, such as *Cannabis sativa*. In their analysis of its traditional use in Ayurvedic medicine, emphasized the growing interest in merging traditional herbal methods with industrial and pharmaceutical applications and providing scientific validation for them[5].

The main contribution of the paper is, to focus on the identify and categorize of the medicinal plant leaves using ML algorithms such as Random forest, SVM, DT, LR and KNN. It specifically uses the leaves such as Neem, Aloe Vera, Papaya, and Pumpkin due to their well-known benefits in skin care, such as treating acne, soothing burns, and enhancing skin glow. By identifying the leaf type and linking it to appropriate skin care treatments, the suggested methodology seeks to close the knowledge gap between traditional herbal remedies and recent machine learning techniques.

After passing an image of a twig Neem leaf, the model correctly recognizes it and responds with "Neem: Treats acne and fungal infections." Aloe Vera pictures prompted: "Aloe Vera: Heals burns and moisturizes skin," and Papaya and Pumpkin outputs represented their roles in exfoliation and skin brightening; respectively. This result also demonstrates the fact of that the system is not a mere classifier but rather is a "smart" assistant that provides context based medical information in the area of skin care it shows in the below TABLE 1.

TABLE 1: Leaf images and its uses

Name of the Image Dataset	Leaf name	Medicinal uses
Image 401	Neem	Used for burns, skin care, and digestion
Image 410	Aloe Vera	Used for acne, skin infections, and diabetes

Image 350	Papaya	Improves digestion, treats dengue, and boosts immunity
Image 480	Pumpkin	Good for eyes, skin, and heart health

The papers next parts are structured as follows: The study that has been done on skin care procedures in leaf images is reviewed in Section II. The leaf dataset's properties are provided in Section III. The several machine learning models used for skin care identification are discussed in Section IV, and the metrics used to evaluate these models' performance are described in Section V, and the work is concluded in Section VI.

II. RELATED WORK

Mardiana et al. used, images of herbal leaves were classified for medical purposes using the VGG16 CNN architecture. Used the small dataset, which only included a few different kinds of herbal leaves, made generalization difficult[6]. Patil & Sasikala et al. presented a weighted KNN method for classifying and recognizing therapeutic herbs. With an accuracy rate of 88.3%, the approach demonstrated satisfactory accuracy and computational efficiency. Nevertheless, overlapping features and noisy data in leaf images were problems for KNN[7].

Bhelkar & Sharma et al. focused on the identifying medicinal plants based on their leaf shape using deep CNNs. One major flaw was overfitting brought on by inadequate real-world augmentation. Seasonal variations and the impact of sickness were not taken into account[8]. Khate & Sharma et al. used a basic neural network to classify different types of medicinal plants. The model was shallow compared to modern CNN designs, while having a mean precision of 85.7%. It was less effective on unseen data because to limited feature learning and a lack of transfer learning. The dataset lacked cross-species diversity and was comparatively limited[9].

Rao et al. identified the leaves of plants with medicinal properties using a deep learning architecture and a special CNN. The model shows good feature extraction with over 90% accuracy. Due to the dataset's lack of environmental noise and lighting variation, its real-world application was limited. Pictures were taken of a few different kinds of plants[10]. Booth et al. used for its potential future applications in optimization, even though its main focus is quantum-accelerated constraint programming. Quantum computing is introduced in order to minimize computational cost in high-dimensional problem domains. The lack of dataset is used in the classification of plants[11].

Bose et al., leaf diseases in medicinal plants were identified using SVM. SVM's performance declined when dealing with complicated, non-linear characteristics in large datasets, despite its respectable accuracy of 87.6%. One constraint of the dataset was the lack of field-based sick samples, and precision was impacted by the imbalance among plant species[12]. P. Karur et al. Kaur and Singh studied the application of ML methods for the categorization of herbal plants by comparing the performance of SVM and KNN classifiers utilizing extracted leaf properties such as shape, texture, and color. When handling intricate feature patterns, their research shown that SVM performed better

than KNN in terms of accuracy and resilience. One major drawback was that the dataset mainly focused on a small number of herbal species, with little class diversity[13].

Pushpanathan, K. et al. used ML methods for medicinal plant identification, Pushpanathan and his team examined a number of algorithms, such as CNNs, SVM, and DT. The study emphasized deep learning's advantages in managing intricate plant picture characteristics and enhancing classification precision[14].

Begue A. et al. used the SVM, KNN, and DT are some of the machine learning approaches. Although efficient, the algorithm had trouble classifying species that were visually similar, which occasionally resulted in incorrect classification. The model's capacity to generalize in real-world scenarios was severely hampered by the use of a small dataset, which included only 30 distinct plant species and comparatively few image samples per class. [15].

III. DATASET DESCRIPTION

A. Summary of the dataset

In the data collection, the Leaf dataset was curated from several reputable open-source platforms, including, mendely, and real-time field data, comprising approximately 2000 images in healthy leaf images format. The dataset utilized in the study was 2000 high resolution images of leafs 500 images per medicinal leaf of four crops viz., Azadirachta indica (Neem), Aloe barbadensis (Aloe vera), Carica papaya (Papaya), cucurbita (pumpkin). The medicinal leaf images are taken from the mendely such as Alovera leaf dataset[16], Papaya leaf dataset[17], Neem leaf dataset[18] and Pumpkin leaf dataset[19]. All images were taken in natural light, and on one uniform plain background to minimize noise and enhance visuals. At this time, a special effort was made to include only images of healthy leaves (avoiding disease symptoms, discolorations or deformations) in order to achieve uniformity. Such a balanced dataset, reduces the risk of class bias problems, thus ensuring reliable training and testing of machine learning-based models. The data records were divided into training (70%) and test (30%) categories, and the images were resized to 450×600 pixels to meet the identification of objects model's input specifications. In the below the Fig.1. shows the sample dataset leaf images.

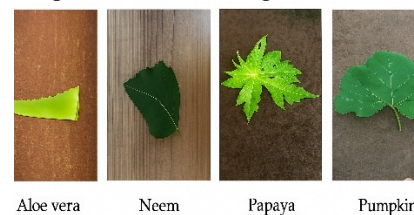


Figure 1 : Sample Leaf Images of Dataset

B. Dataset Description

Neem, aloe vera, papaya, and pumpkin are all equally represented among the 2,000 photos of medicinal plant leaves in the dataset. 20% went toward testing, and the remaining 80% was allocated to training. Assuring more consistent outcomes, this method assisted in assessing the model's performance on various data subsets. SVM, RF, DT, LR, and KNN were among the machine learning models that

were trained with feature vectors that were taken from the preprocessed photos. Using grid search and cross-validation, hyperparameter adjusting was carried out to determine the optimal setup for every algorithm shown in the TABLE 2.

TABLE 2 DETAILS OF TRAINING, AND TESTING IMAGES USED

Name of the leafs	Training	Testing
Neem	380	120
Aloevera	300	200
Papaya	350	150
Pumpkin	400	100

C. Preprocessing

To improve the model's accuracy and predictability in identifying the leaves of the medicinal plants neem, aloe vera, papaya, and pumpkin, a number of data augmenting approaches were used to the leaf images. They were scaling, cropping, flipping, random rotation along vertical and horizontal directions and color, brightness, contrast and saturation augmentation. These adjustments allowed the model to learn from images taken under all kinds of conceptions and lighting conditions. The dataset due to the lack of ability to overfit to the too clean training data. In general, the model's ability to extract important features from the leaves and accurately predict their uses was increased using regularisation. Then the image size is transformed to resized(image,(128,128)).The markings were manually checked for their consistency and accuracy, however, some failure mechanisms were noticed especially with low light conditions and complex background. These classifications may serve as potential areas of refinement. While the current system is dealing with leaf treatments, the framework is generic and could also be applied to scouting and monitoring of a field to find healthy leafs for treatments and their utilization in medicinal context.

IV PROPOSED METHODS

To facilitate our classification, This method performed a feature extraction phase which transformed the raw images into a structured representation. All these features together represent the visual texture and morphological information of the leaf. The medicinal leaf classification system's suggested architecture aims to replicate expert-level plant identification and associate each leaf with its associated dermatological benefit. After that, the image goes through a preprocessing step where tasks like labeling, grayscale conversion, and resizing are carried out. By removing noise, these procedures help get the data ready for insightful analysis. The image then goes through feature extraction using methods such as Local Binary Patterns (LBP) to extract texture information, Color Histograms to record color distribution patterns, and Histogram of Oriented Gradients (HOG) for shape and contour recognition. The distinct visual traits of every leaf are reflected in these extracted features. To make sure all features contribute equally during the learning process, feature scaling through standardization is applied prior to model training. A range of machine learning classifiers, including LR, DT, RF, KNN, SVM, and others,

are then fed the scaled features. The four selected medicinal plants pumpkin, papaya, aloe vera, and neem are then differentiated from one another using the attributes that were extracted. To attain the best accuracy and generalizability, training is improved through the use of cross-validation and hyper parameter tuning. It was this hand-crafted feature vector that served as the input to all machine learning models, allowing for the decision boundaries to be learned accurately and the Fig. 2 shows it is proposed architecture whether what all are the process is performed in the model.

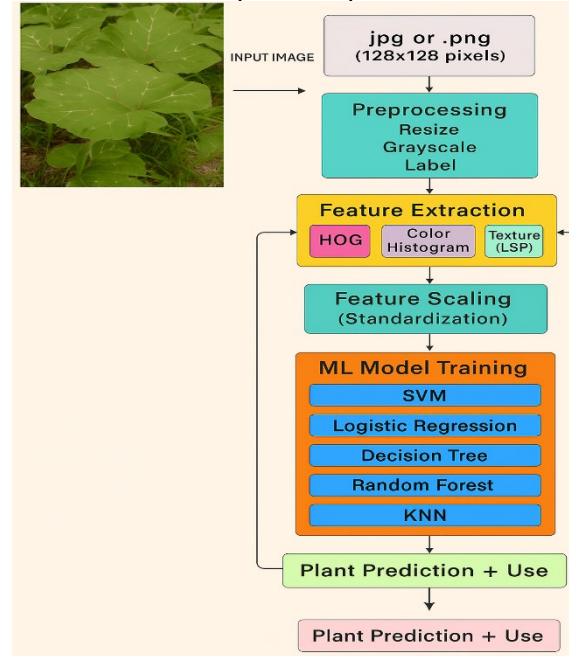


Figure 2:Proposed Architecture

In order to develop a system that can automatically recognize medicinal herbs from their leaves and propose their benefits for the skin, this research combines modern facilities in machine learning techniques with traditional plant based knowledge . Data augmentation methods like flipping, rotation, brightness adjustment, and contrast changes were used to improve the dataset. This guarantees that the models can effectively generalize even in a variety of environmental circumstances . There is some strength in each approach. SVM is optimal in high dimensional space with clear margin, whereas RF combines ensemble learning for better generalization. Interpretable rule-based structure of DT, Probabilistic modeling in a simple way of Logistic regression and Instant-based learning of KNN. In the training phase, 80% of the datasets were chosen and the 20% of the test datasets were tested.

A.SUPPORT VECTOR MACHINE

For high-dimensional feature classification problems, such as leaf identification, SVM is the perfect supervised learning technique. In this project, handcrafted features taken from the textures, edges, and forms of leaves were used to train SVM. In this instance, it creates an ideal hyper plane that maximally divides the classes of neem, aloe vera, papaya, and pumpkin. When used in conjunction with

the right kernels, it excels at handling non-linear boundaries. It demonstrated good classification abilities throughout training when the features were clearly different. It made it possible for the model to accurately generalize to unseen leaf samples under actual lighting and environmental conditions.

B. RANDOM FOREST

An ensemble learning approach called RF constructs several DT and combines their results to increase accuracy. RF managed a variety of leaf properties for this project, such as intricate textures and shifting color gradients. It was the most reliable model in operation and worked remarkably well, with the greatest accuracy of 92.75%. By averaging predictions across trees, Random Forest minimized overfitting and guaranteed stable classification. It demonstrated great generalization to various contexts and operated efficiently with the enriched dataset. Random forests feature significance capabilities also helped discover which visual features contributed most to predictions. It was perfect for real time plant recognition with skincare application coupling because of its predictive reliability.

C. DECISION TREE

A rule-based technique called a DT divides data into subsets according to feature thresholds in order to classify the data. In the present research, a treelike model was constructed using characteristics like color, shape, and leaf edge structure. It was helpful for visualizing how each plant kind was distinguished from the traits because of its interpretability. Decision Tree allowed quick decision-making and achieved excellent accuracy despite being simpler than RF. It demonstrated how some characteristics like the smooth edges of aloe vera were more important in categorizing specific leaves. The model was simple to use and produced results that could be explained, making it a good tool for training or instruction. Before using more intricate models, Decision Tree helped to confirm that feature extraction was consistent.

D. K-NEAREST NEIGHBORS

KNN is a slow learning algorithm that classifies new input points by utilizing its closest training samples. When leaves were images with uniform lighting and orientation, it performed well and showed strong neighborhood clustering. KNN provided respectable performance with reasonably accurate classification and was easy to apply. It did not, however, perform as well with overlapping characteristics in complex leaves and was susceptible to noise. Its accuracy of 73.75% suggests that it could be improved in more variable settings. Nevertheless, KNN helped measure the efficacy of other models and offered a helpful baseline performance.

E. Logistic Regression

LR is a probabilistic method of classification that models the probability of belonging to a class. In this research, LR produced the likelihood that an image belonged to each plant class using the sigmoid and softmax algorithms. On the balanced dataset, it worked well, especially in feature space

where the classes were linearly separable. Because of its excellent pre processing, LR was able to attain a good 90% accuracy even though it was simpler than other models. It was appropriate for lightweight installations in mobile or embedded devices due to its speed and cheap computational cost. By offering a probabilistic interpretation, LR made it simple to connect its results to medical use cases. In comparative evaluation, it supported more intricate algorithms like RF and SVM by acting as a reliable baseline model.

V. RESULT AND DISCUSSION

A. Performance metrics

Accuracy, precision, recall, and F1-score were the four primary performance metrics used to evaluate the model. These measures provided a complete summary of each model in terms of the accuracy of plant leaf images classification and handling of the imbalanced class distribution.

Accuracy: The proportional of datasets with correctly labeled skin care leaf images. This statistic is frequently used to evaluate a classifier's overall performance.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}$$

Precision: The proportional of leaves that are actually healthy out of all those that have been identified. This parameter is a measure of the classifier's ability to avoid false positives.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall: The proportional of real, healthy leaf images in the collection that are accurately recognized. This statistic assesses how well the classifier avoids producing false negative results.

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-score: A weighted average of recall and precision that provides a single evaluation of the classifier's overall effectiveness. When recall and precision are both significant, this statistic can be effective.

$$F1 - \text{score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

TABLE 3: Comparison of testing and training

Algorithm	Precision	Recall	F1-score	Accuracy
Logistic Regression	0.91	0.90	0.90	90.00%
SVM	0.90	0.88	0.88	87.50%
Decision tree	0.88	0.88	0.88	87.75%
Random forest	0.90	0.94	0.92	92.75%
KNN	0.81	0.74	0.70	73.75%

From the TABLE 3 it Successfully recognize the model are correctly identified as TP (true positive), other objects that are (incorrectly) wrongly marked, other objects that are

correctly identified from TN (true negative), and peacocks (false negative) due to FN (FALSE negative).

VI.CONCLUSION

This examine suggests how well system studying models RF specifically classify the leaves of therapeutic leaves such as aloe vera, neem, papaya, and pumpkin with the accuracy of 92.75%.The Random Forest algorithm gives a best accuracy compared to other algorithms in a leaf images for skin care treatment. In addition to recognizing plant species from leaf images, the version buddies each one with its own set of dermatological and fitness advantages, as enumerated within the plantuses dictionary. These advantages include healing burns, pimples, and improving heart and pores and skin health. The gadget's capability as an clever herbal advice tool for clients, researchers, and the natural cosmetics area is highlighted by means of its twin reputation and suggestion abilities.By adding more species of medicinal plants and gathering field photos in real time under various environmental circumstances, the dataset can be enlarged.To make the gadget very accessible for useful dermatological advice, a cell app or internet-based totally platform that lets in customers to snap a photo of a leaf and right away acquire its call and skin care benefits can be created.

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